Autonomous Driving Agent for Automated Manual Transmissions (AMTs)

Hierarchical State Machine and PID Control for Non-Linear Vehicle Dynamics

The Need for Automated Manual Transmissions (AMTs)

AMTs combine the raw mechanical efficiency of a traditional manual gearbox with the ease of an automatic.

They offer superior fuel economy and direct power transfer compared to traditional fluid-driven torque converters, while being cheaper to manufacture and maintain.

Because they use physical clutches and synchronizers instead of fluid couplings, they require a highly intelligent electronic agent to execute shifts smoothly without stalling the engine or jarring the passengers.

The Engineering Challenge

Mathematically replicate human driving intuition for discrete AMT actuators.

- Balancing clutch slip, throttle input, and gear selection dynamically.
- Compensating for exponential aerodynamic drag and variable gravitational slopes.
- Preventing engine stalling, shift shock, and target velocity overshooting.

Agent Architecture: Macro & Micro Control

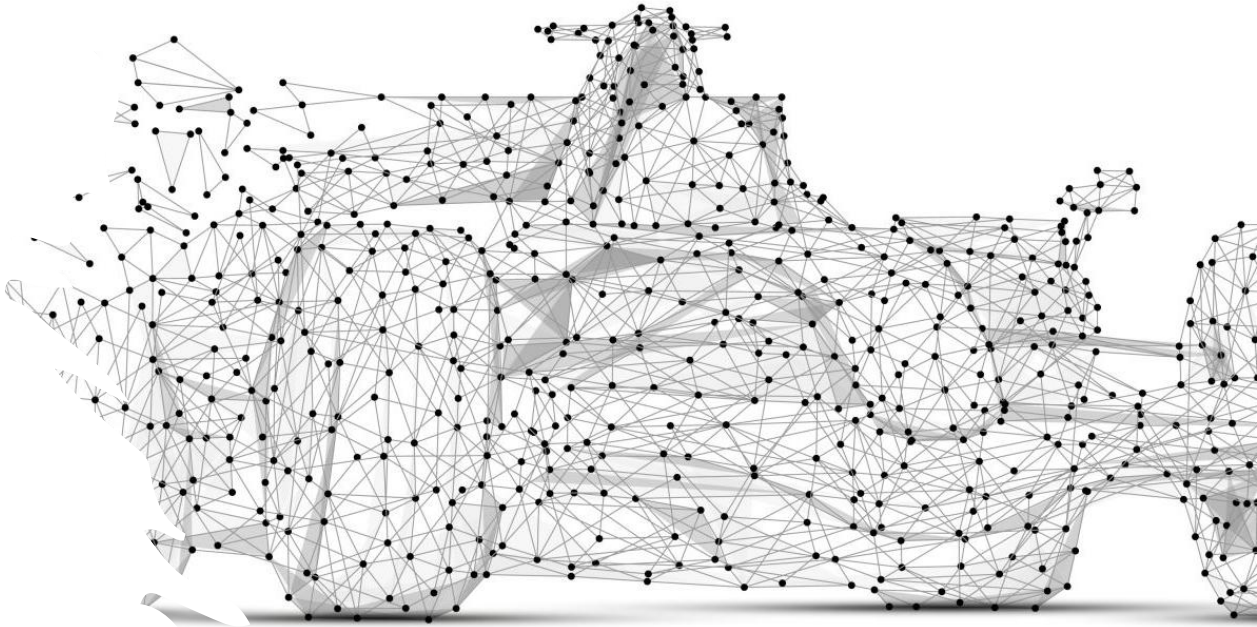
The Problem: Managing high-level driving behaviors (like launching or stopping)

while simultaneously requiring precise, continuous pedal adjustments.

The Solution: A dual-layer control architecture.

•**Finite State Machine (FSM):** Orchestrates the macro-states (Idle, Launching, Driving, Shifting) safely.

•**PID Controllers:** Manage the precise, micro-level application of the throttle and brake actuators.



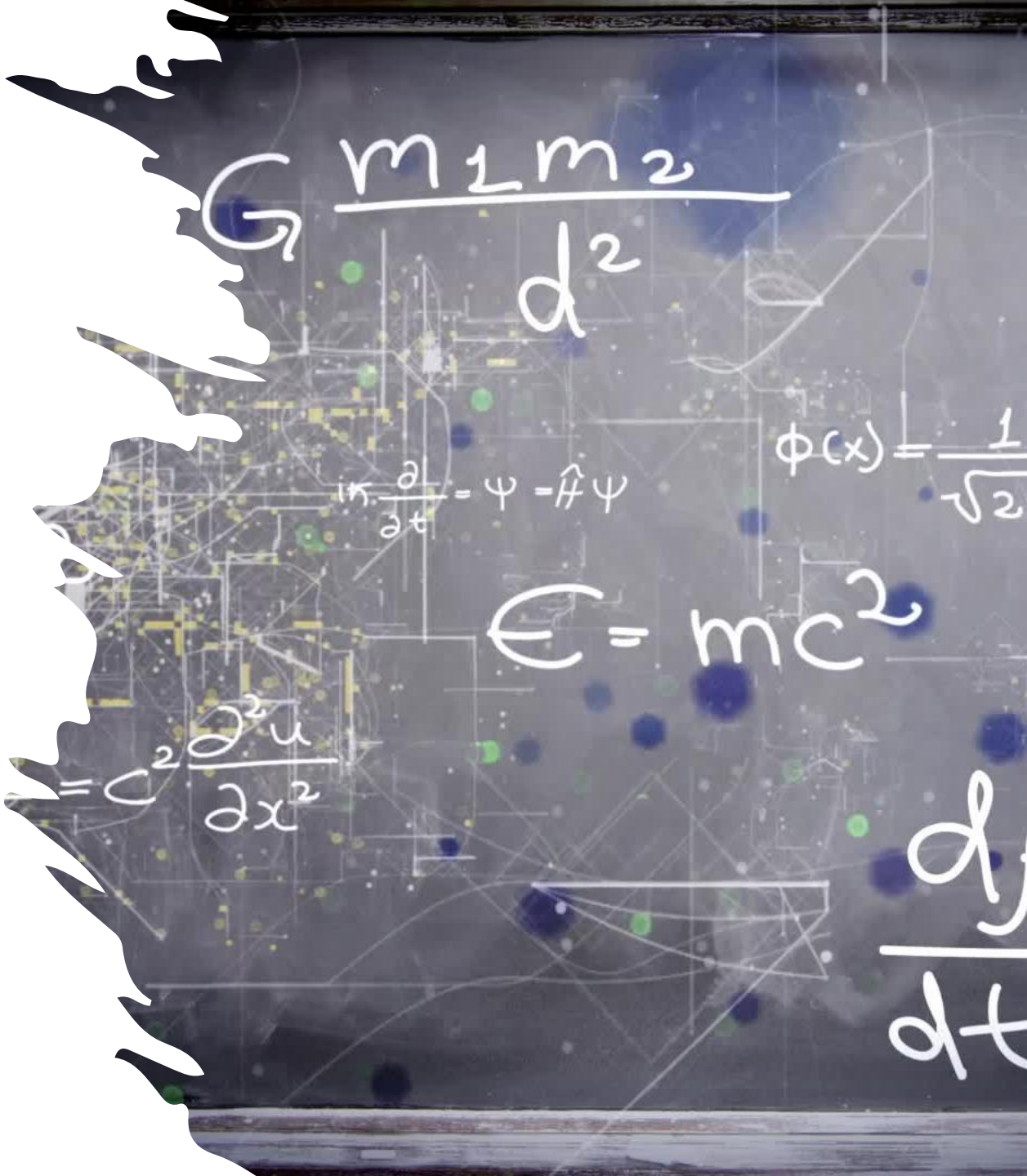
Intelligent Gear Selection

The Problem: Rigid, pre-programmed shifting tables fail on steep hills,

causing erratic "gear hunting" and boundary oscillation.

The Solution: Real-time physics evaluation and objective scoring.

- The agent calculates the exact physical force required to overcome gravity and drag.
- Available gears are mathematically scored based on engine power limits.
- A hysteresis penalty is applied to non-active gears to eliminate rapid, unnecessary shifting.



Speed Stabilization & Safety

The Problem: Long accelerations cause the controller memory to "wind up"

and overshoot target speeds, while downhill descents cause simultaneous pedal battling.

The Solution: Mathematical constraints and mutual exclusion.

- **Anti-Windup Clamping:** The PID memory is mathematically capped to strictly prevent speed overshooting.
- **Descent Management:** Software locks prevent pressing the gas and brake at the same time. The agent relies on engine braking first, engaging physical brakes only if a safe coasting buffer is exceeded.